



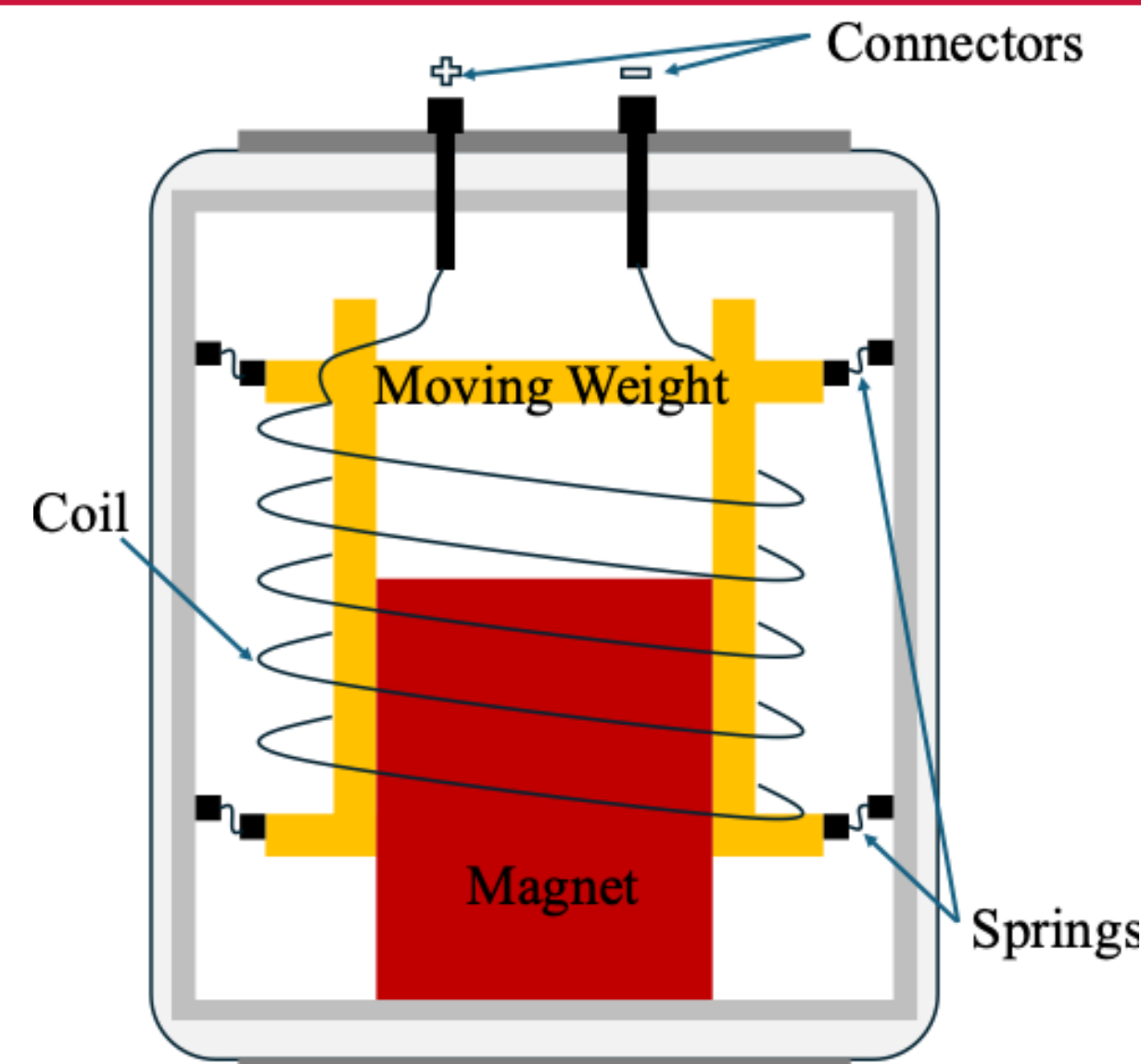
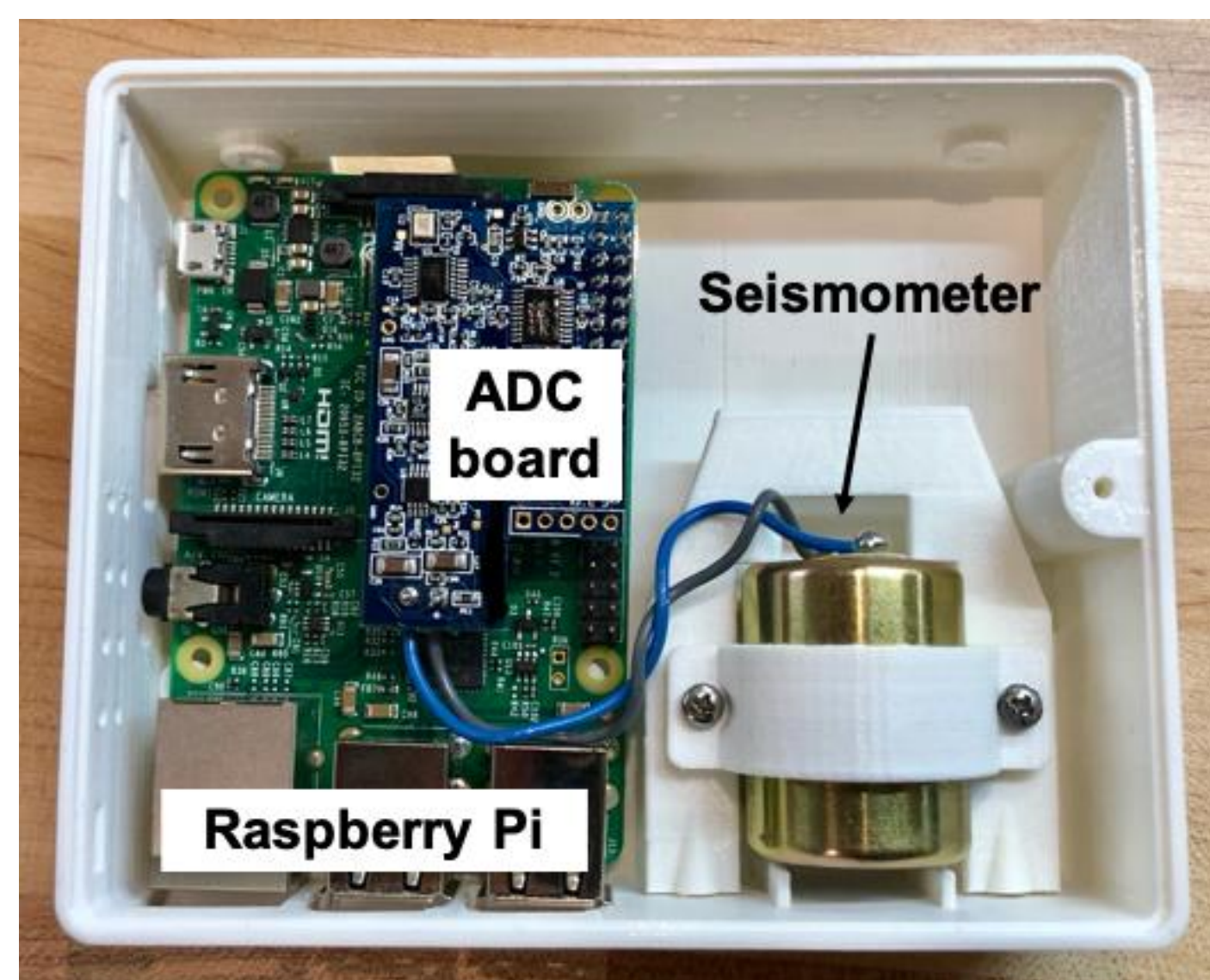
UNIVERSITY OF
GEORGIA

School of Electrical and
Computer Engineering

SeatVitals: Geophone-Based Unobtrusive Vital Sign Monitoring for Office Environments

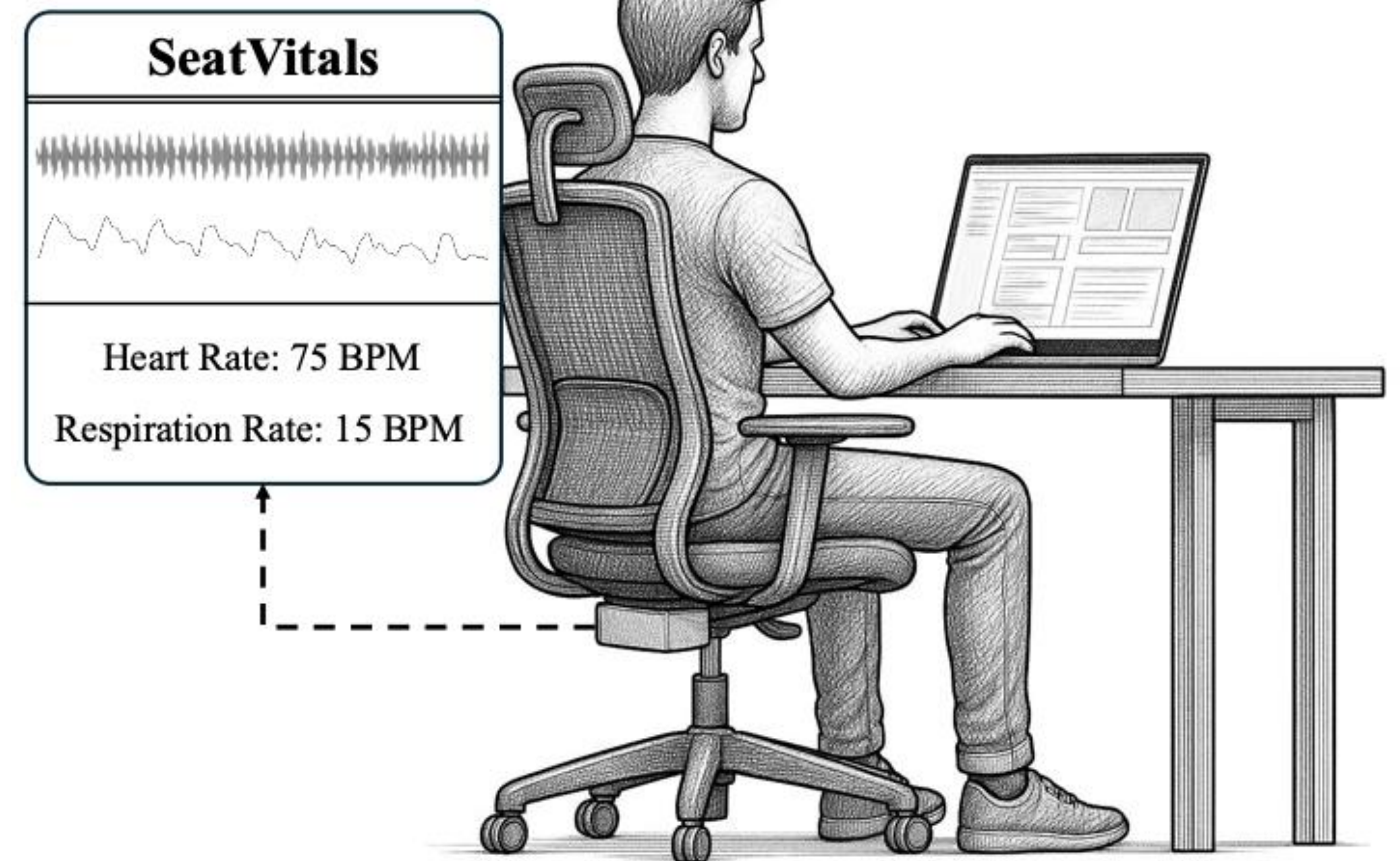
Yingjian Song*, Jiayu Chen*, Yida Zhang, Zixuan Zeng, Zaid Farooq Pitafi, Bradley G. Phillips, Xiang Zhang, Zaipeng Xie, Fei Dou, Qin Lu, Wenzhan Song (wsong@uga.edu)

System Details

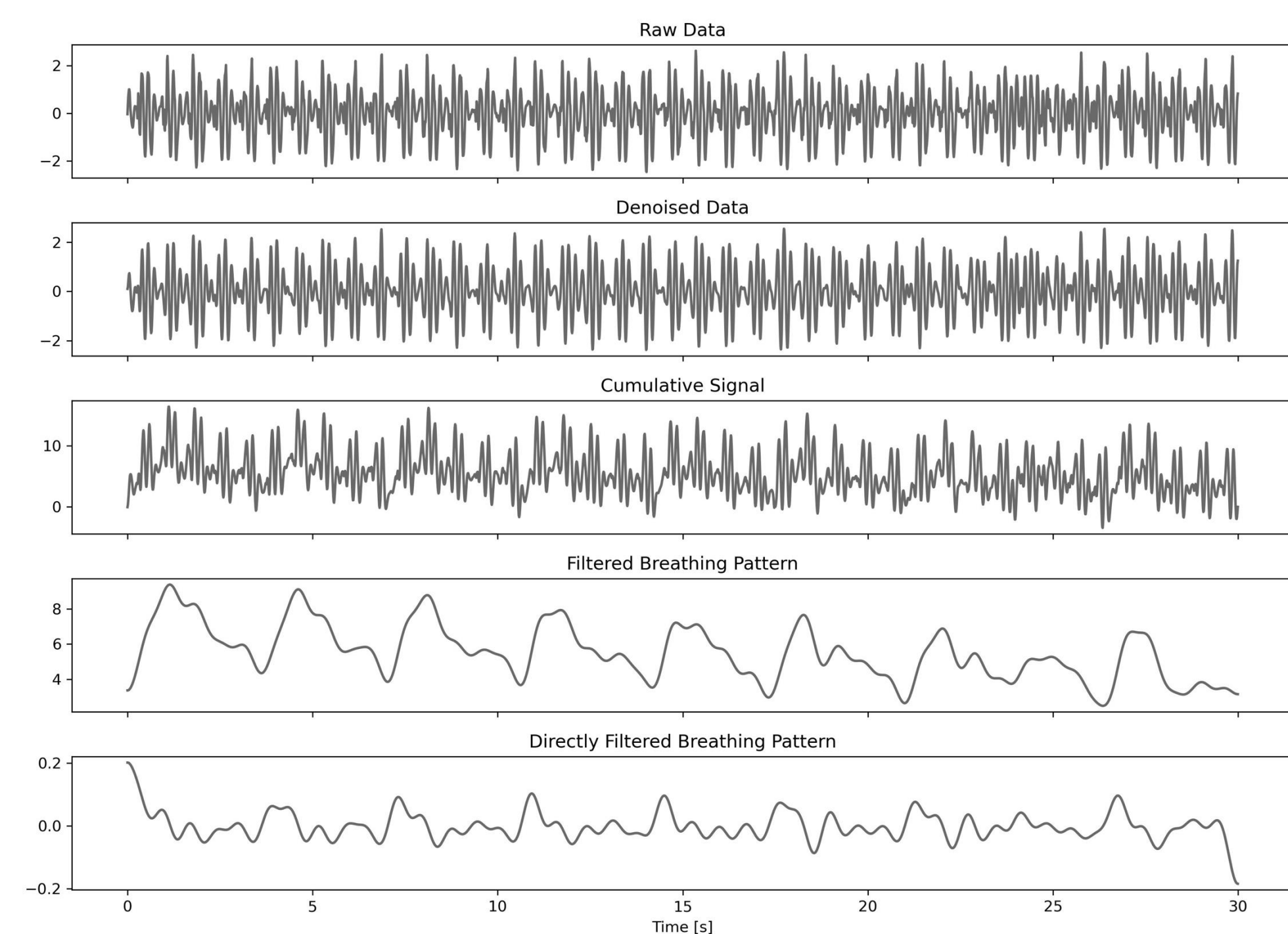


SeatVitals consists of a Pi serving as the onboard computer, an ADC with a 100 Hz sampling rate, and a vertical geophone. The geophone is capable of capturing subtle microvibrations generated by human cardiac activity.

System Illustration



Framework Overview



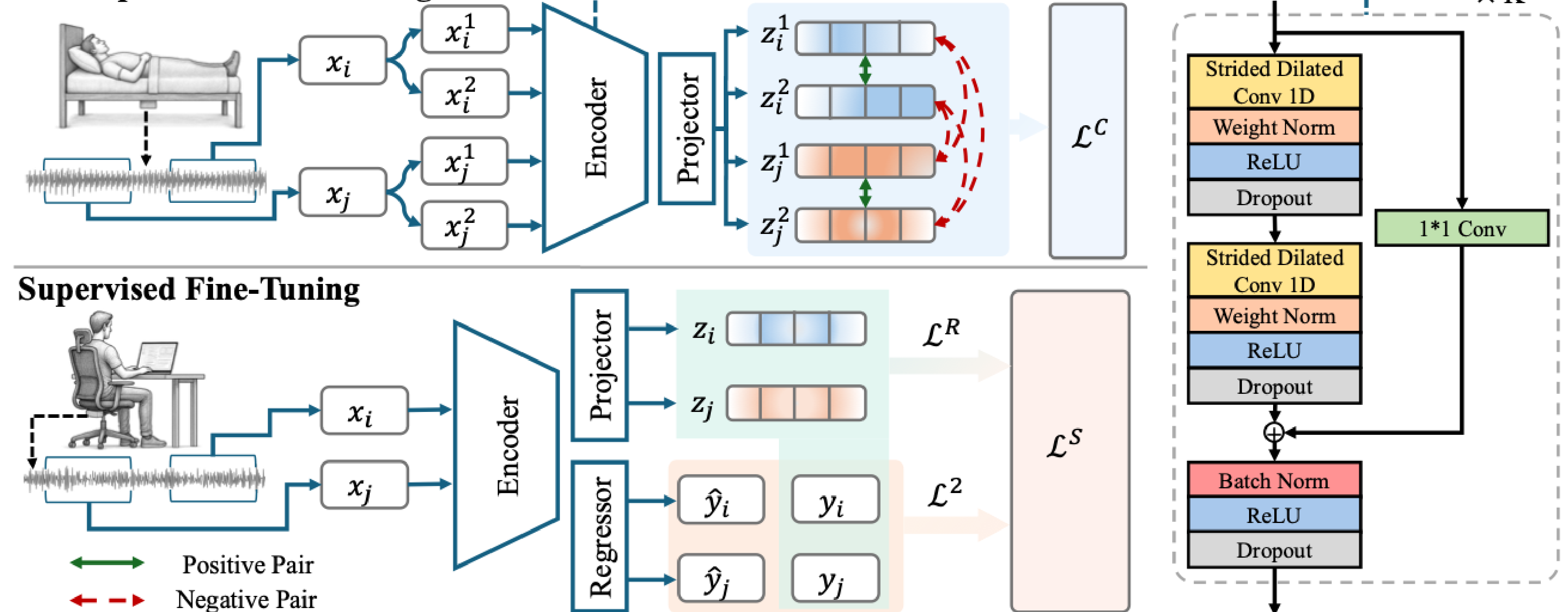
First row: raw data lacking clear morphological landmarks, with continuously varying waveforms.

Second row: pre-processed signals for heartbeat patterns.

Third and fourth rows: pre-processed signals for respiration patterns.

Last row: result of directly applying a lowpass filter to the raw data.

Self-Supervised Pre-Training



Self-Supervised Pre-Training

We augmented views of samples are processed by a shared encoder, and the resulting representations are further mapped by a projector into the contrastive embedding space. A contrastive loss is then employed to encourage higher similarity between embeddings derived from the same input, while reducing similarity with embeddings from other inputs.

Supervised Fine-Tuning

During the supervised fine-tuning phase, the objective is to strengthen the association between learned features and their corresponding labels, ensuring that representations capture task-relevant characteristics.

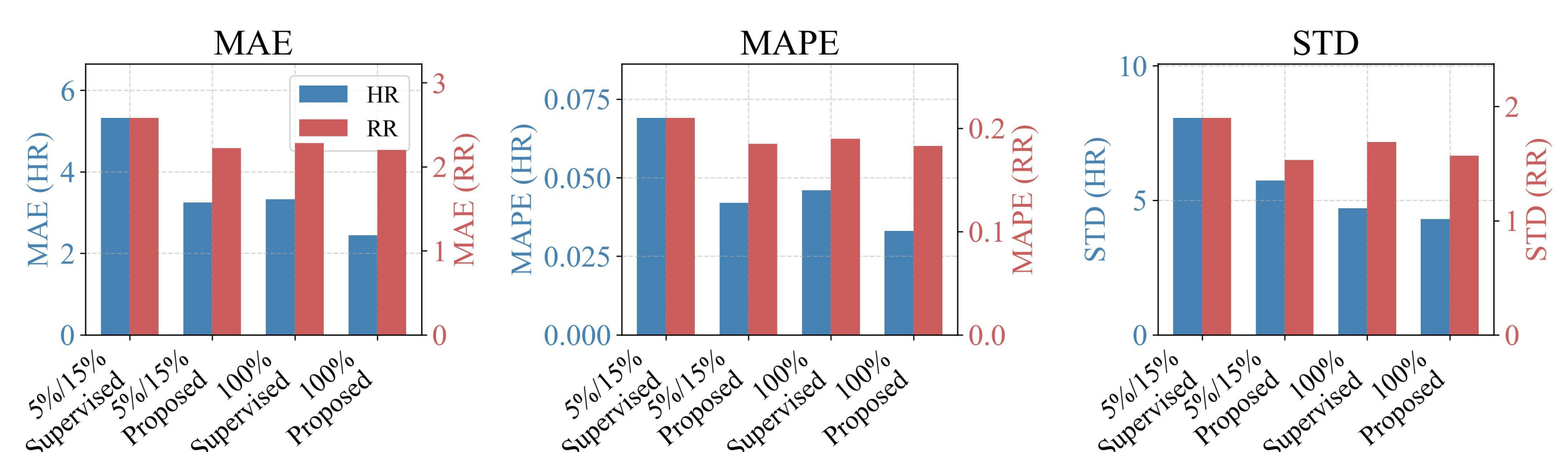
Experiments

Performance Comparison against Baselines

Table 1: Performance comparison with baselines. Bold indicates best performance per row.

Method	Vital	MAE ↓	MAPE ↓	STD ↓
HB-Phone	HR	7.12	0.103	17.06
	RR	4.48	0.342	3.36
Yingjian Song	HR	3.88	0.050	8.82
	RR	3.37	0.243	2.83
SUPERVISED	HR	3.33	0.046	4.71
	RR	2.28	0.190	1.69
PROPOSED	HR	2.44	0.033	4.30
	RR	2.20	0.183	1.57

Evaluation under Limited Labeled Data



The pretrained model consistently outperforms both baselines under the same amount of labeled data, with a more pronounced advantage when labeled data are scarce.